GROUND_SIZE 1000
GROUND_COLOR 104,160,70
SKY_COLOR #1A2230
WAIT
WAIT
SETCOLOR color=#FF444G mode=Solid

VF (.vf) Language Specification

1. General Definition

• Name: VF

• File Extension: .vf

• Comment Syntax: #

• Purpose: Domain-Specific Language for Voxel Forge Code Studio.

• Execution Model: One command per line, case-insensitive, space-separated arguments, key=value named options, quoted strings for spaces.

2. Core Data Types

• Number: Integer or Float

• Boolean: true/false, yes/no, on/off, 1/0

• Color: Hex (#RRGGBB) or RGB (0-255)

• URL: http:// or https:// only

3. Scene Commands

• GROUND_SIZE – Set ground dimensions

• GROUND_COLOR – Set ground color

• SKY_COLOR – Set sky color

• CLEAR – Clear scene modules

4. Timeline Commands

• AT – Absolute time cursor

• WAIT – Move timeline cursor forward

• ROTATE – Scheduled rotation (axes, speed, duration)

• SETCOLOR – Change color mode or value

5. Web & Asset Commands

• URLCHECK – Validate link before usage

• IMAGE – Convert image URL into mesh

• MODEL – Load .vf3d model

• PICTURE – Store image as asset or mesh

• AUDIO – Store validated audio metadata

• VIDEO – Store validated video metadata

6. Primitive Mesh Commands

• BOX – width, depth, height
• SPHERE – radius, segments, rings
• CYLINDER – radius, height, segments
• FACEMODULE – Multi-face image-based 3D module

7. Class & Modular System

• CLASS – Define reusable blueprint
• NEW – Create module instance
• SET – Modify properties
• APPLY – Attach module to scene
• DELETE – Remove module definition

8. Module Properties

• Common: type, color, tx/ty/tz, rx/ry/rz, scale
• Box: width, depth, height
• Sphere: radius, segments, rings
• Cylinder: radius, height, segments

9. Script Management

• SAVEVF – Save script file
• LOADVF – Load script into editor
• HELP – Print command list

10. Target System

• primary
• secondary
• add
• extra
• auto

11. Architectural Characteristics

• Line-based execution structure
• Explicit named properties
• Reusable modular design
• Integrated URL validation

• Timeline-based scene control

• Performance-aware geometry configuration

Official Developer Certification

Name: Laith Abu Halawa

Role: Founder and Architect of VF Language

Statement: This document certifies that Laith Abu Halawa is the original designer and developer of the VF (.vf) scripting language, created for Voxel Forge Code Studio to control procedural 3D generation, asset validation, reusable module systems, and timeline-based scene management.

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